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ITECH3102 – Data Visualization
Week 5

Lab Exercises – Simple data visualizations

ID: _____Name: _____

In this project, we discuss how to use the basic approaches to visualize simple data. Here visualizing simple data refers to that visualization can be achieved through **directly using existing fields** from raw data source. In other words, we do not need to take **simple computation** on the **existing fields** or to combine the existing fields in order to make new dimensions or measures that don't exist in raw data source. When visualizing simple data, in particular, we use basic dot chart, histograms, line chart or scatter plot, etc., to visualize data.

Project 1: Basic bar charts

Identifying your questions

Data visualization normally doesn't start from dataset itself, but rather it starts from questions. Who is your audience? What stories can it tell? What does your audience do with your visuals results?

For example, for given data, you might want to ask the following questions:

How many customers did each store serve?
How much energy did each wind farm produce?
How many patients did each doctor see?

To answer the questions like how much and how many, you, in each above case, need to make a comparison (among stores, wind farms, or doctors) in terms of some quantitative measurement (number of customers, megawatts of electricity, and patients). In this case, bar chart is most likely one of good choices.

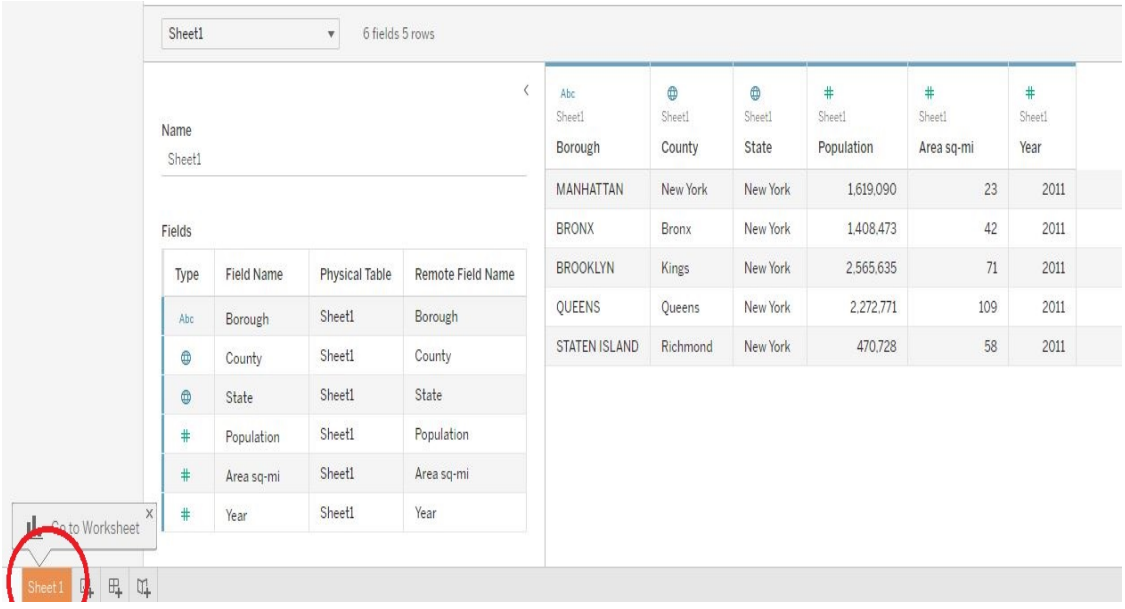
Bar chart

The bar chart is one of the oldest and simplest chart types. It is this simplicity that makes it so appealing. Visualized as bars, measures can be easily understood without much explanation. Comparisons can readily be made: it's relatively easy to gauge the approximate difference in the lengths of two or more bars.

Download Dataset: simple-data-bogoughs.xlsx from week 1's lab in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly
Step 2 Create a bar chart

Step 1. **Load the data into Tableau** and check that your data has been loaded correctly

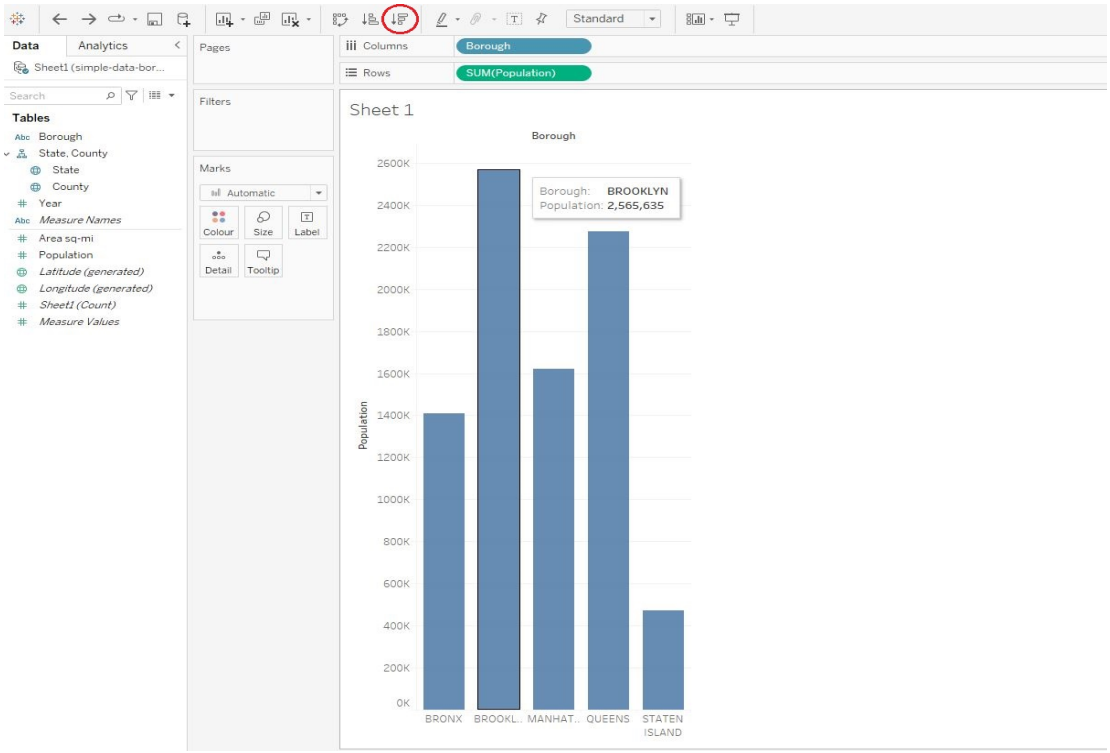


After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

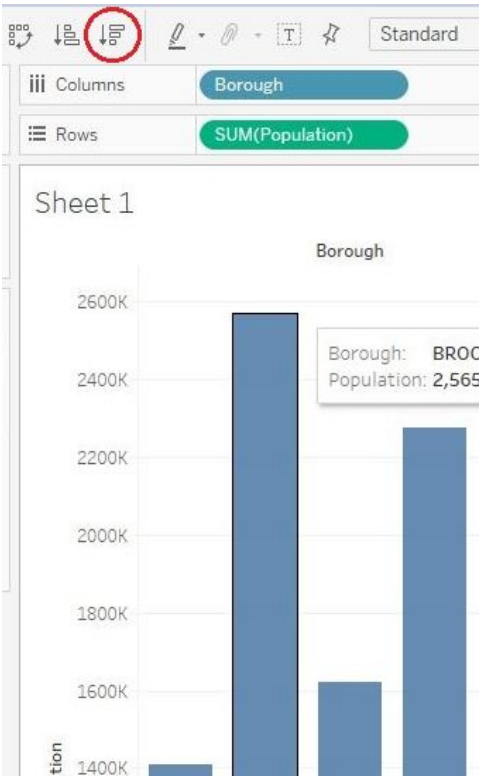
Step 2. Create a bar chart

Drag and drop **Population** field from the Measures section in the data pane on to the Rows shelf. You now have a bar chart with a single bar representing the sum of population for all of the data in the data source.

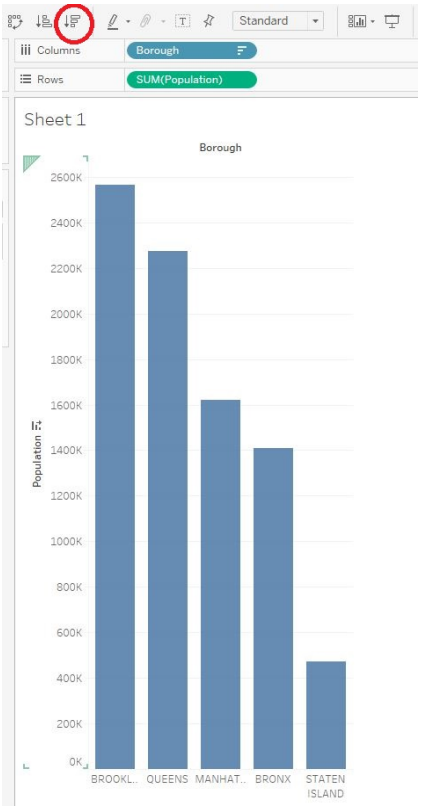
Drag and drop **Borough** field from the **Dimension** section in the data pane on to Columns shelf. This slices the data to give you five bars, each having a length that corresponds to the sum of population for each area.



Step 3. Sort the bars

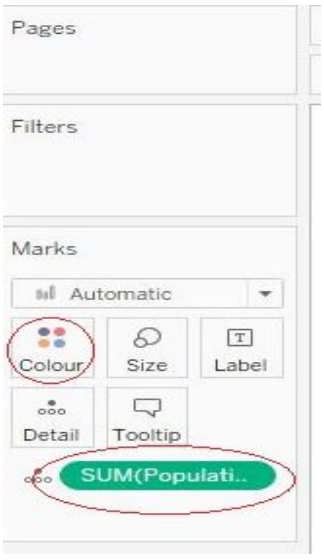


You can simply click the Sort icons in the toolbar above the canvas to sort the bars by length. After clicking the Sort icons, you'll get:

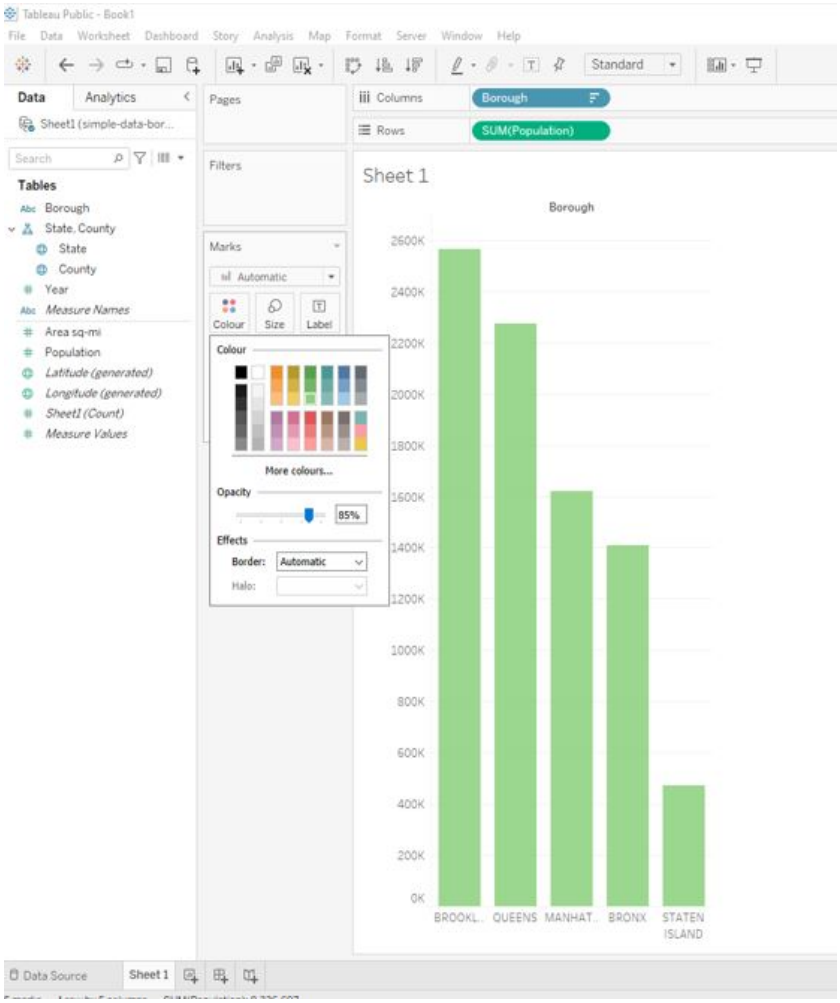


Step 4. Change color

Firstly, Drag and drop **Population** to Color on Marks card.



Then click **Color** to edit color and then change color to green.



Project 2: Dot charts

Project 2 Dot chart

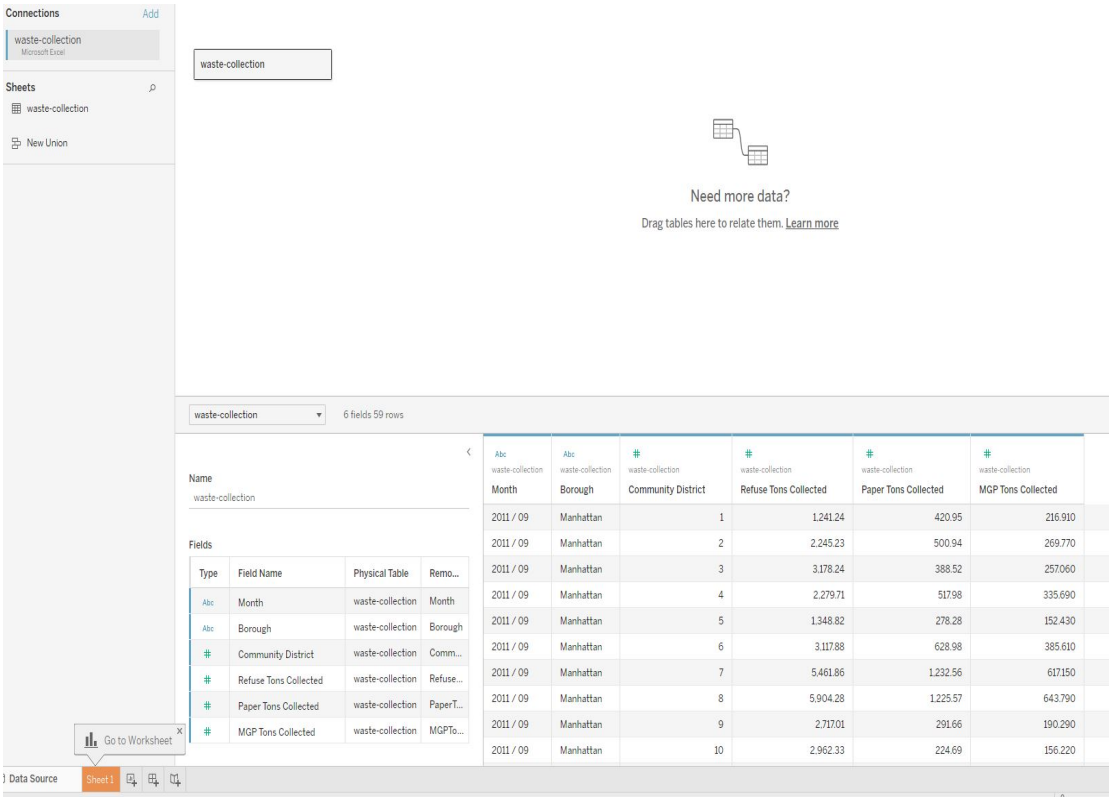
Dot chart is a simple way of showing a change or a range of data across multiple dimensions, especially when comparing two dimensions. It is a good way to show quantities because it leverages the strength of human visual system to compare quantities based on a symbol's position.

Task: The given dataset provides information about population and area in a city. The goal of this task is to be familiar with basic data types, visual encoding, and operations by using chart and map.

1. Download Dataset: waste-collection.xlsx from Week 5 in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly.
Step 2. Create a dot chart.

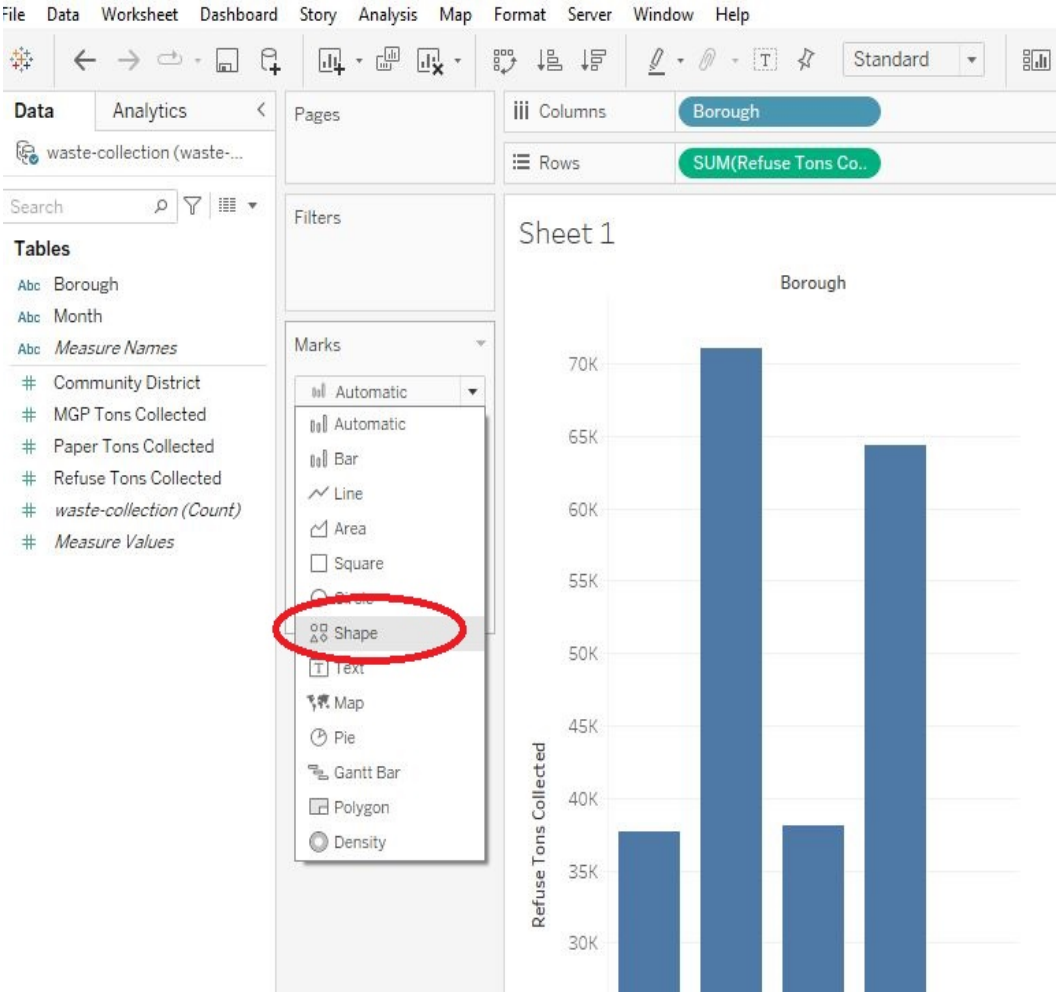
Step 1. Load the data into Tableau and check that your data has been loaded correctly.



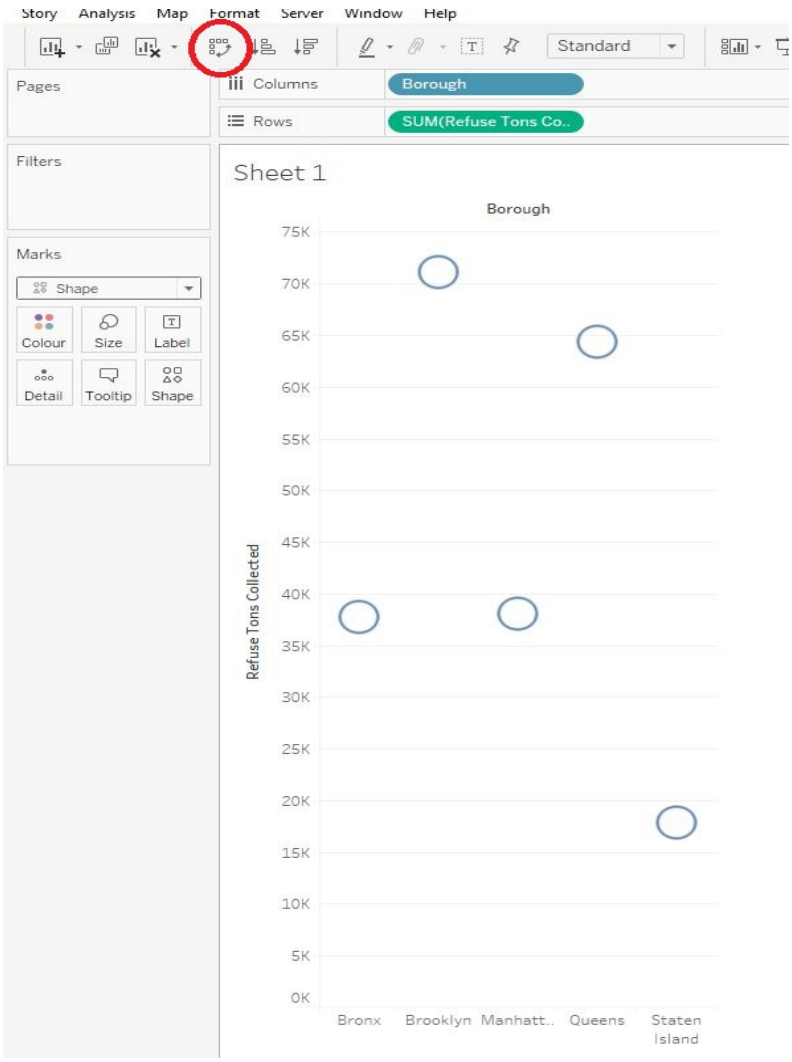
After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create a dot chart.

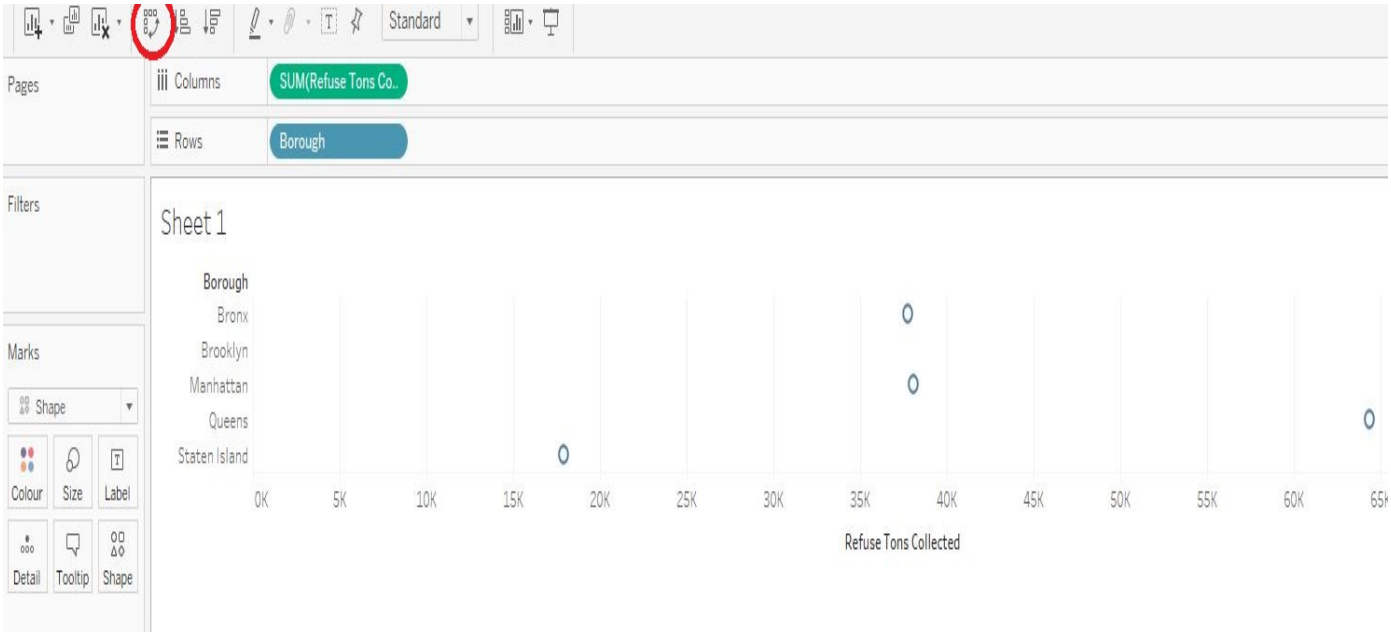
Drag and drop **Refuse Tons Collected** to Row shelf, drag and drop **Borough** to Columns shelf, and this will create a bar chart.



From Marks card, click Automatic and select Shape, which will create a dot chart.

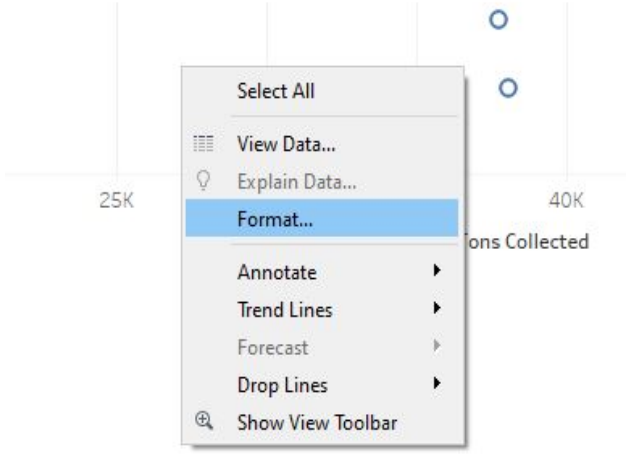


Click the Swap tap from tools bar to rotate dot chart.

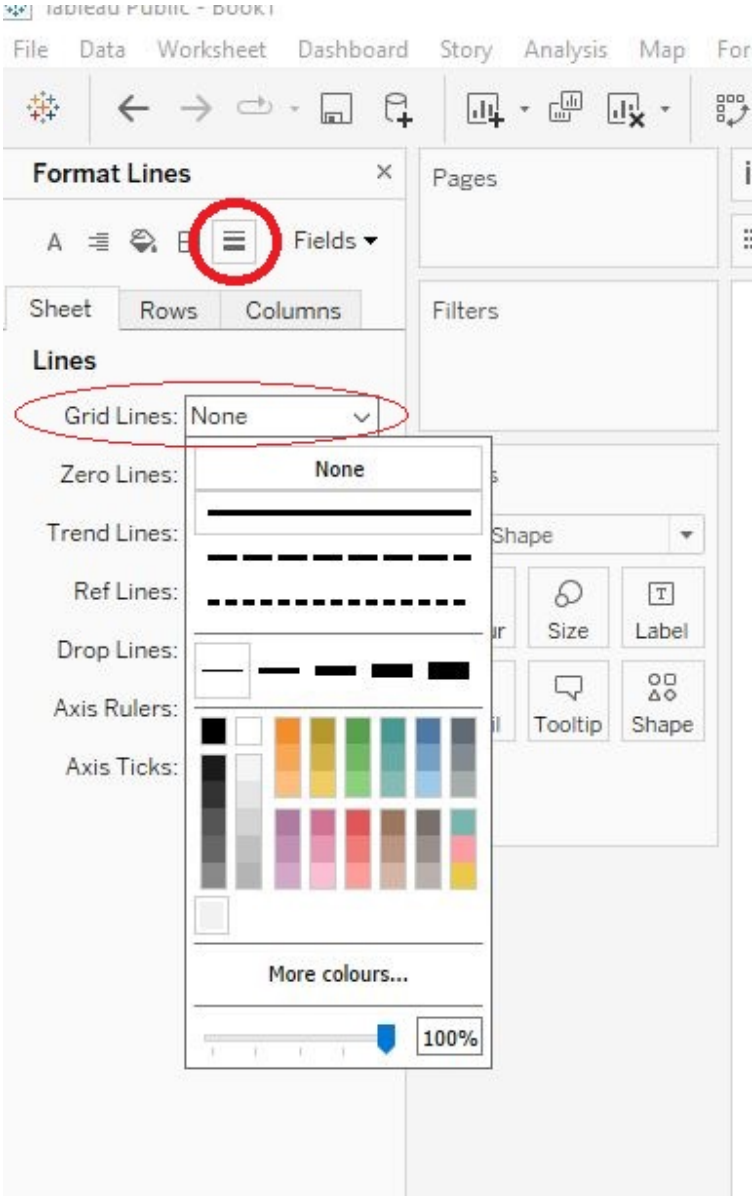


Finally, we'll add grid lines to form the dot chart.

Right click anywhere in the dot chart and select Format.



After clicking Format, Data pane on the lefthand changes to Format Pane. Then click Lines tap and select a line type from the Grid Lines drop-down selector.





Project 3: Line chart

Changes over time are often best visualized with a line chart. Line charts are very useful for looking at trends over time. A line chart is mainly used when you are comparing data over time. The dots on a line chart are typically connected with straight lines. Tableau also offers the option to use step lines and jump lines, instead.

Dates can be split into Discrete and Continuous. As mentioned in the introduction, a discrete field will create headers, whereas a continuous field will create an axis. When using Dates, a discrete date will be using an individual date part, which is a standalone element like Months or Weeks. When using continuous dates, it will return the date value. When looking at Months, Tableau will return the month and the year instead of just the month.

Straight Lines

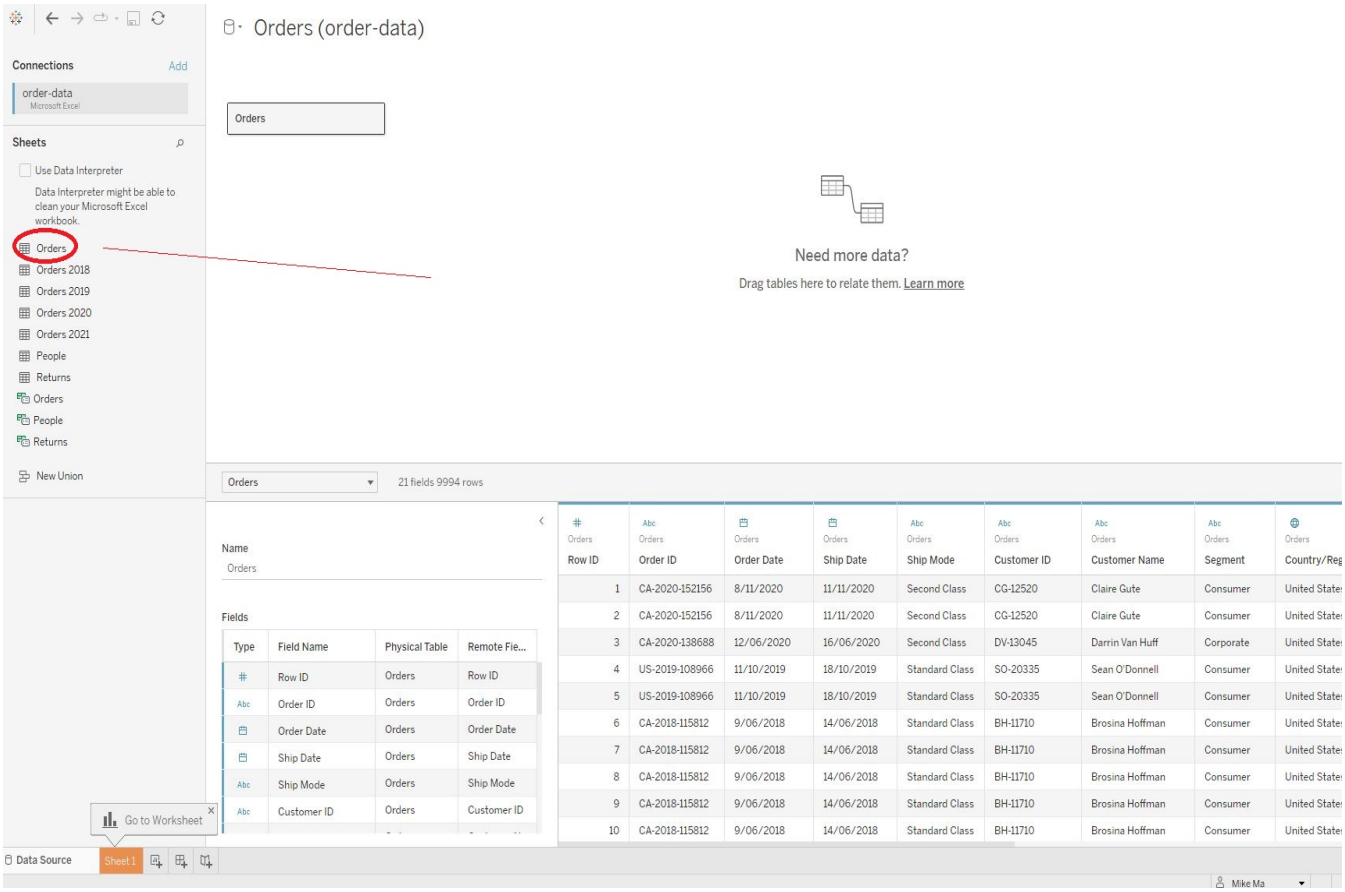
Straight lines are the most commonly used type of line charts and are suitable for showing temporal developments and trends.

Task: The given dataset provides information about population and area in a city. The goal of this task is to be familiar with basic data types, visual encoding, and operations by using chart and map.

1. Download Dataset: order-data.xlsx from Week 5 in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly.
Step 2. Create a line chart.

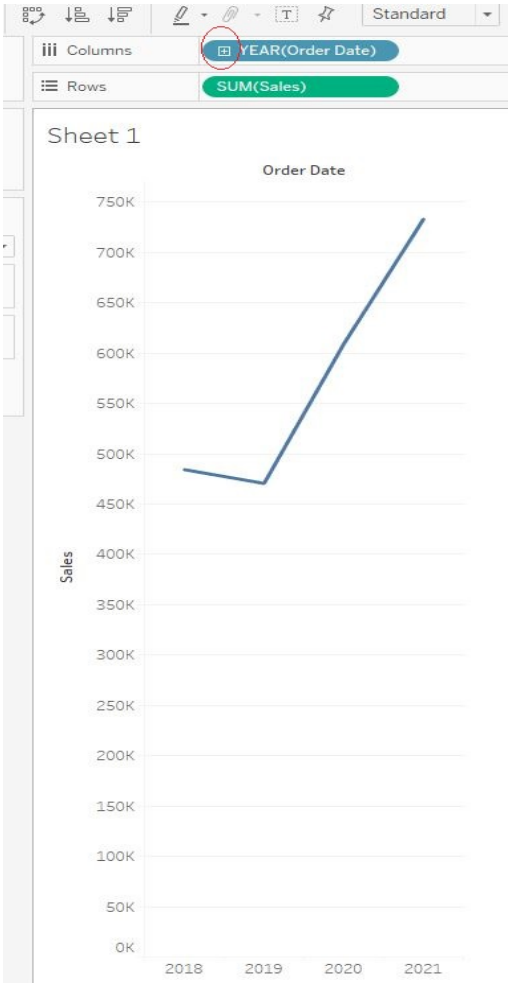
Step 1. From the lefthand Data pane, drag table Orders to the canvas and drop it. Check that your data has been loaded correctly.



After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create a straight line

Drag and drop **Order Date** to Columns shelf, and drag and drop Sales **to** Rows shelf, and this will create a line chart.



Click the plus sign next to YEAR (Order Date), you can break down the data further to quarter level and get the following chart:

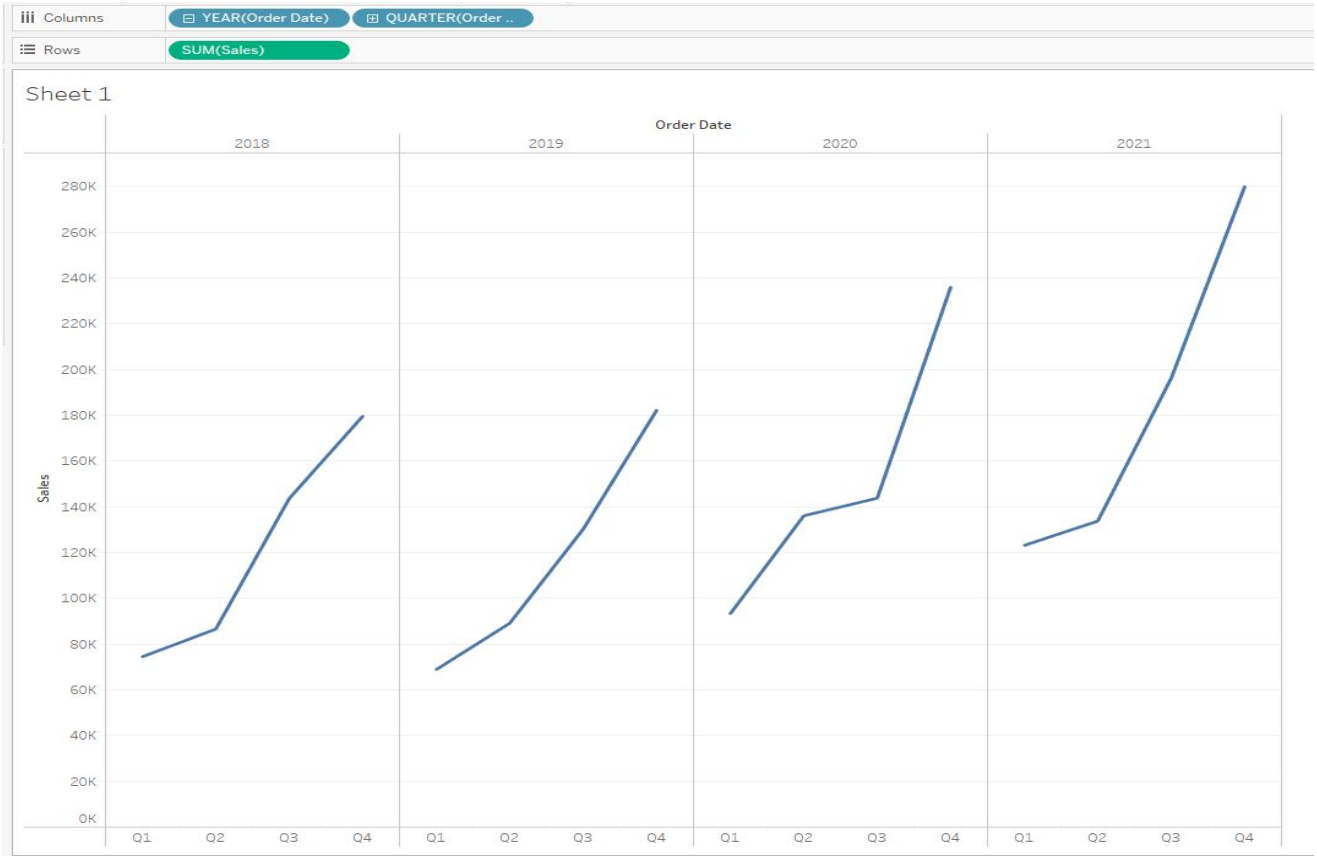
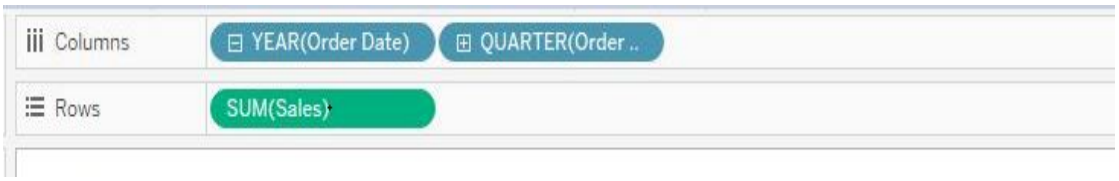
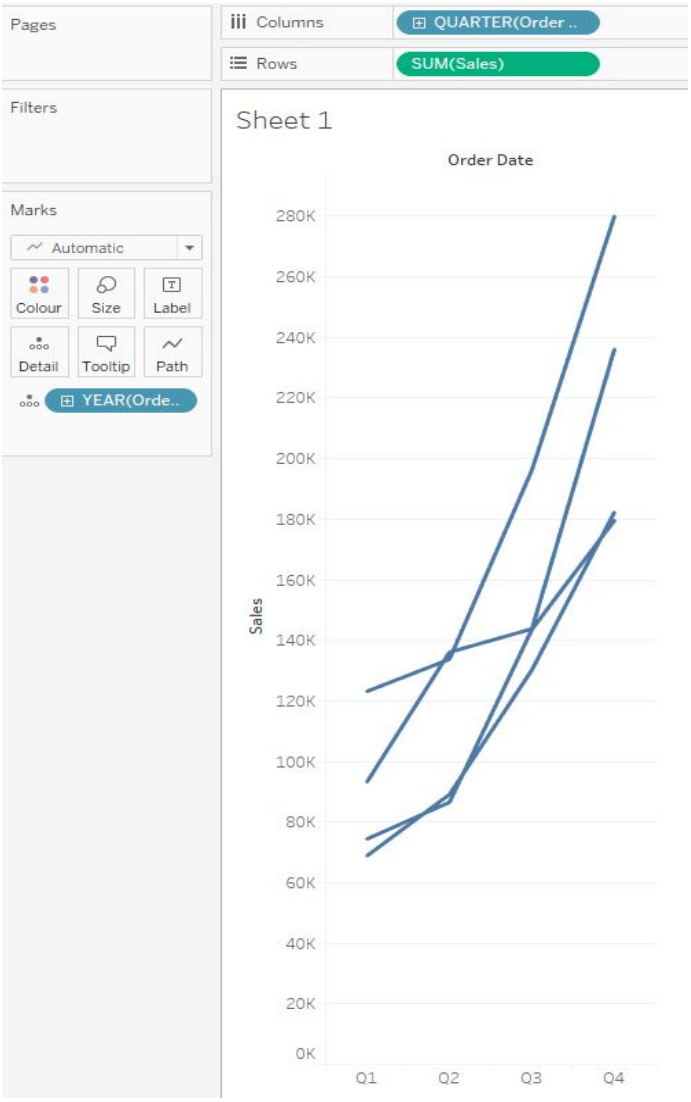


Tableau automatically treats dates such as Order Date as hierarchies. From the following figure, we can see that the highest level of the hierarchy is year, and then quarter.

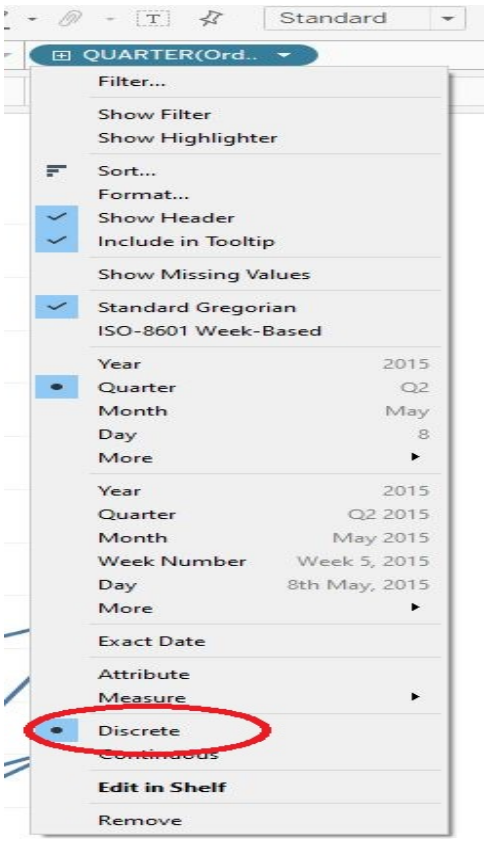


Step 3. Compare the value of a quarter in different years

To more easily compare the values of a quarter in different years, we can move YEAR (Order Date) from Columns to Color on the Marks card, like the following:



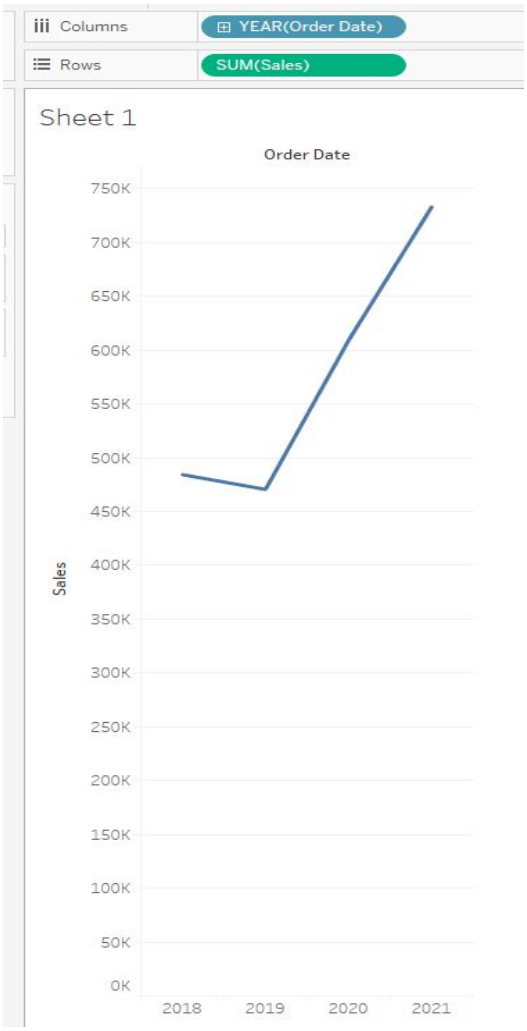
By default, Tableau treats date fields as discrete dimensions. So, if you click QUARTER(Orders) drop-down selector, you can see that quarters are discrete fields.



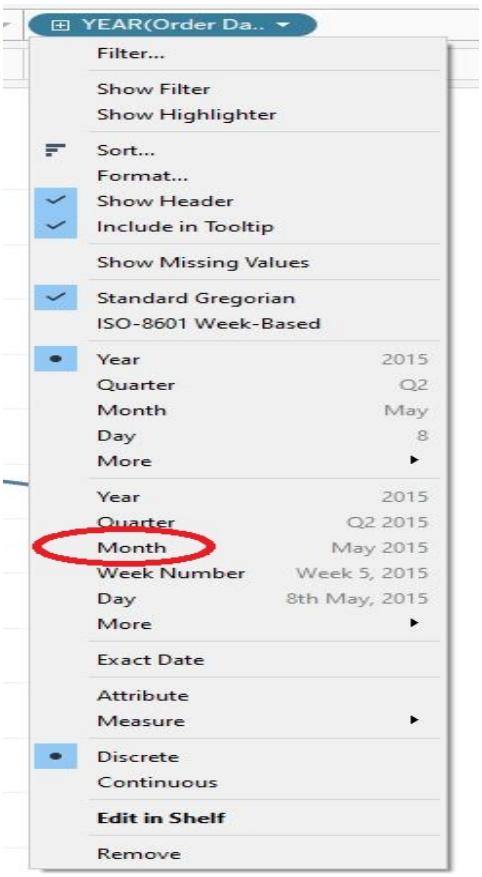
Step 4. Use continuous line charts to see trend

Now if we intend to check the full range of the trend over time such as looking the total trend of sales by month, continuous line chart is one of good choices.

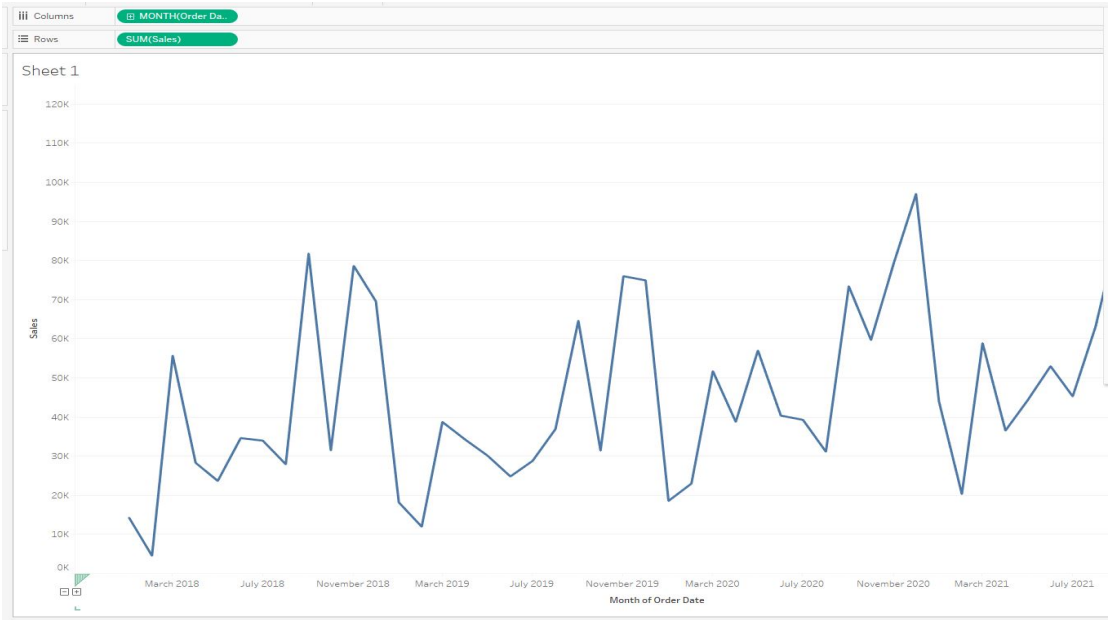
Drag and drop Sales to Rows shelf, and drag and drop Order Date to Columns shelf, you'll get a line chart:



YEAR (Order Date) is highlighted as blue color. If you click YEAR (Order Date), you can see that Year is a discrete field on columns.

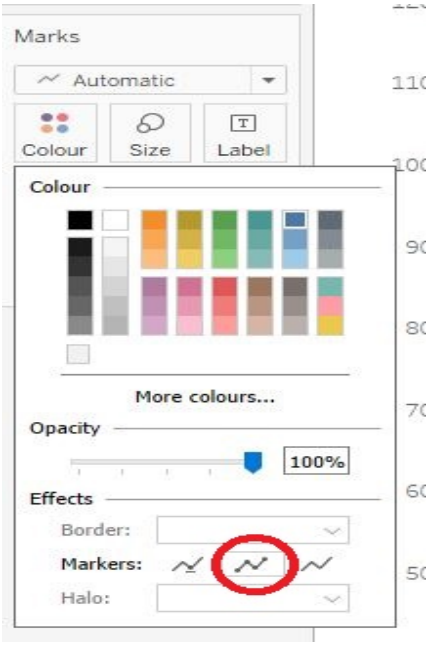


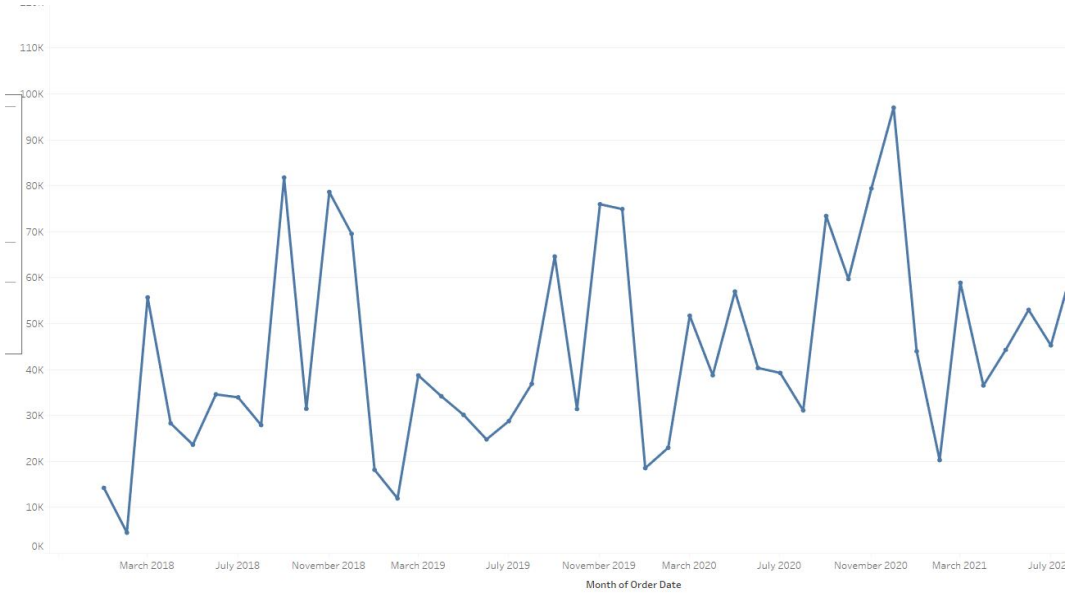
We now select Month and get the following continuous line chart. MONTH (Order Date) is now highlighted as green color, which indicates that it is a continuous field.



Step 5. Add markers to the line

Click color on the Marks card and select related markers that are added to the line.





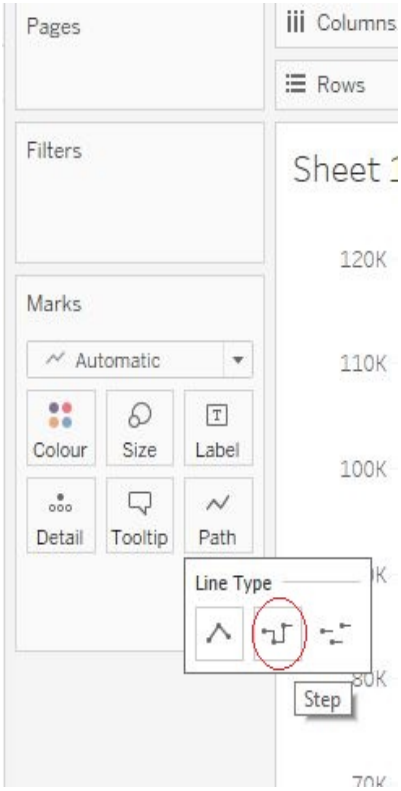
Step 5. Highlight significant changes between the data points

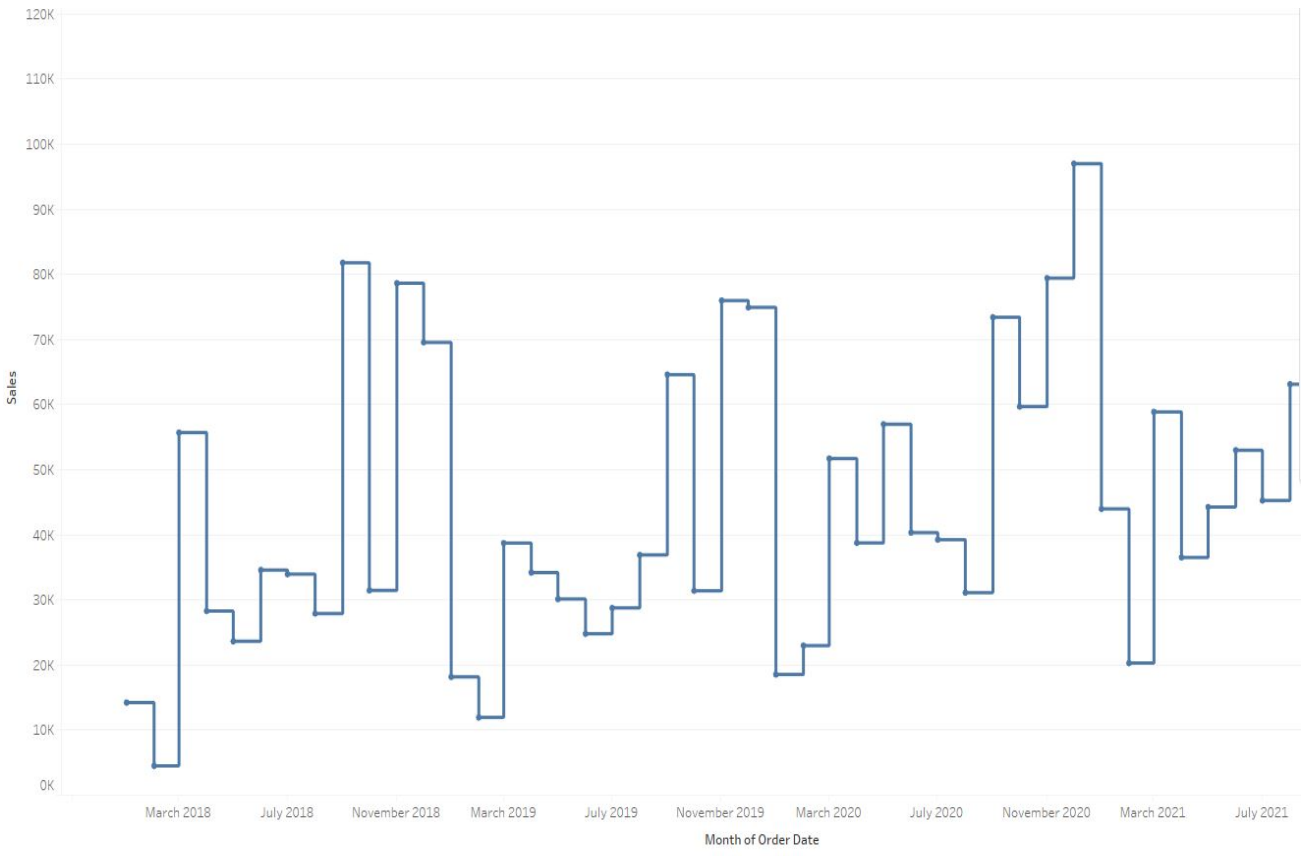
When using the Line mark type, Tableau provides you with several choices by which a line is drawn. Basically, Path in Marks card provides three options: Linear (default), Step, and Jump charts.

(1) Step charts

Step charts are often used to show short-term variations in interest rates.

Click the Path button on the Marks card and then select the line type Step Lines.

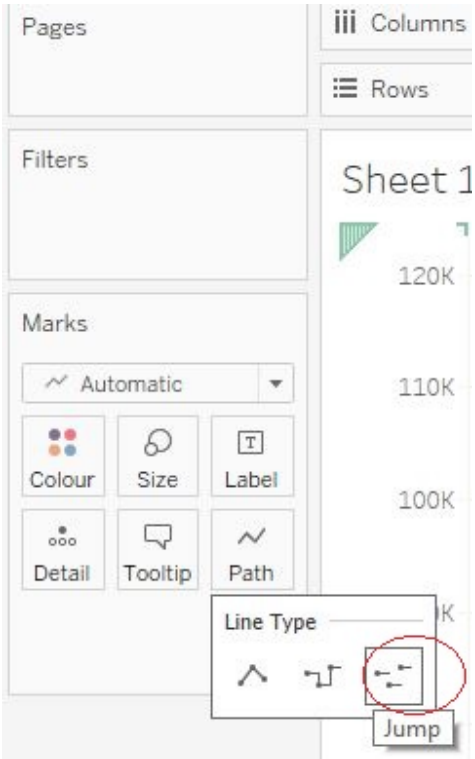

Pages
Columns
Rows
Filters
Sheet 1
120K
110K
100K
Marks
Automatic
Colour
Size
Label
Detail
Tooltip
Path
Line Type
Step
80K
70K

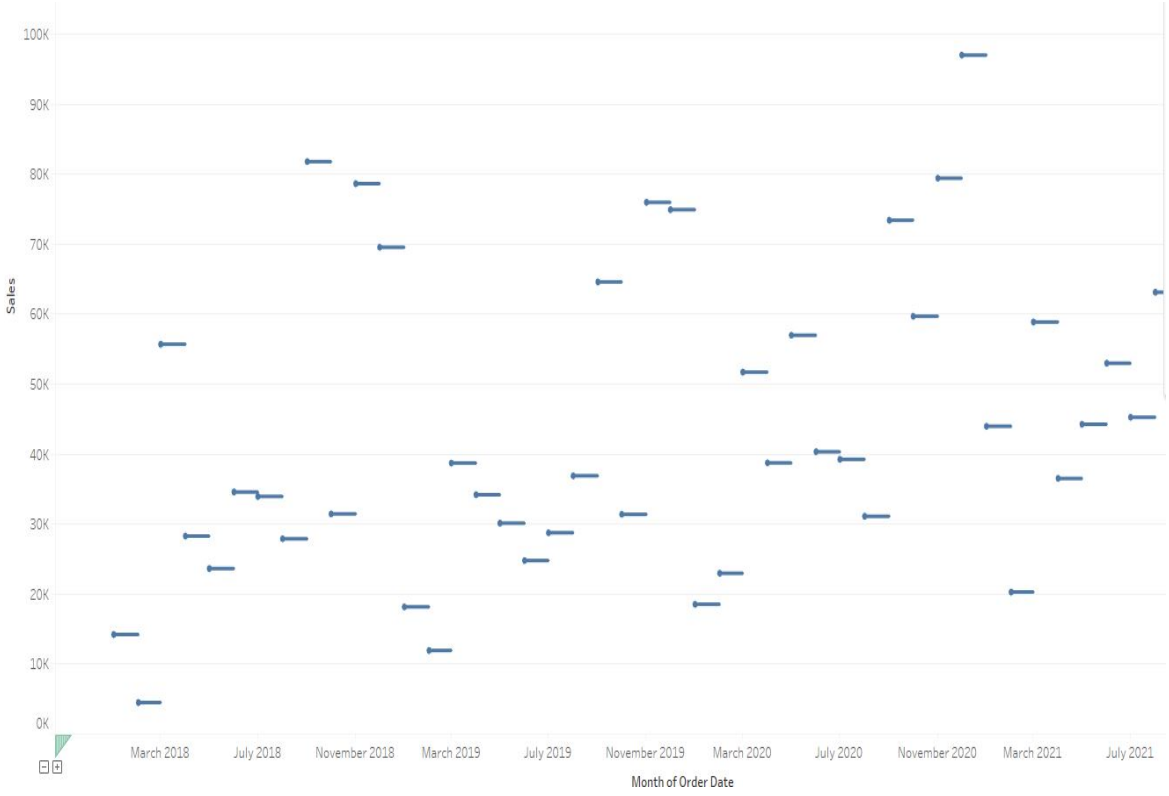

120K
110K
100K
90K
80K
70K
60K
50K
40K
30K
20K
10K
0K
Sales
March 2018
July 2018
November 2018
March 2019
July 2019
November 2019
March 2020
July 2020
November 2020
March 2021
July 2021
Month of Order Date

(2) Jump lines

Jump charts are often used to show the gaps between data such as time-series data containing gaps. For example, jump charts are often used to visualize sensor data, such as data coming from the Internet of Things.

Click the Path button on the Marks card and then select the line type Jump Lines.





Project 4: Create histogram

A histogram chart can be used to show the distribution and frequency of data.

Problem

For example, for given data, you might want to ask the following questions:

- How much does each customer spend at our stores and how does that compare to all other customers?
- How long do most of our patients stay in the hospital? Which patients fall outside the normal range?
- What's the average life expectancy for components in a machine and which last more than average? Are there any components with extremely long or extremely short lives?

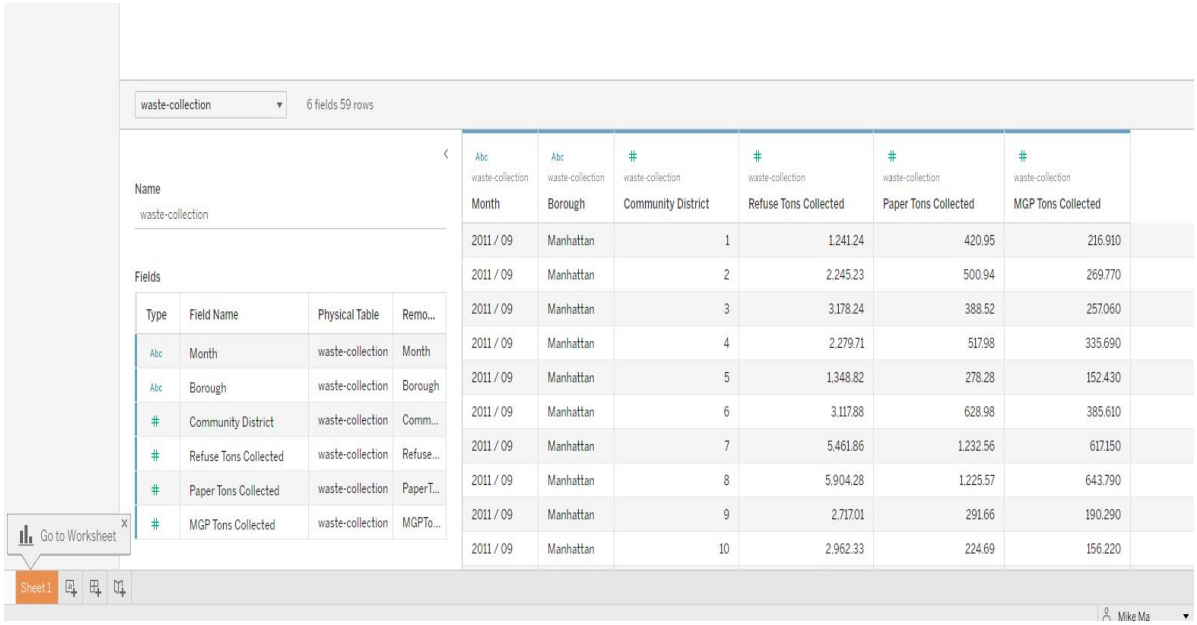
To answer the questions, you, in each above case, seek an understanding of how individuals (customers, patients, components, students) relate to the group. In this case, using histogram might be a good way to do this.

A histogram looks similar to a bar chart, but the bars show a count of occurrences of a value. For example, in this example, borough is an independent variable and we want to show how many community districts yielded certain amounts of refuse.

1. Download Dataset: waste-collection.xlsx from Week 5 in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly.
Step 2. Create a histogram chart.

Step 1. Load the data into Tableau and check that your data has been loaded correctly.



Month	Borough	Community District	Refuse Tons Collected	Paper Tons Collected	MGP Tons Collected
2011 / 09	Manhattan	1	1,241.24	420.95	216.910
2011 / 09	Manhattan	2	2,245.23	500.94	269.770
2011 / 09	Manhattan	3	3,178.24	388.52	257.060
2011 / 09	Manhattan	4	2,279.71	517.98	335.690
2011 / 09	Manhattan	5	1,348.82	278.28	152.430
2011 / 09	Manhattan	6	3,117.88	628.98	385.610
2011 / 09	Manhattan	7	5,461.86	1,232.56	617.150
2011 / 09	Manhattan	8	5,904.28	1,225.57	643.790
2011 / 09	Manhattan	9	2,717.01	291.66	190.290
2011 / 09	Manhattan	10	2,962.33	224.69	156.220

After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create a histogram

Select RefuseTonsCollected field from the table on the left Data pane. Click Show Me and then click the histogram chart.

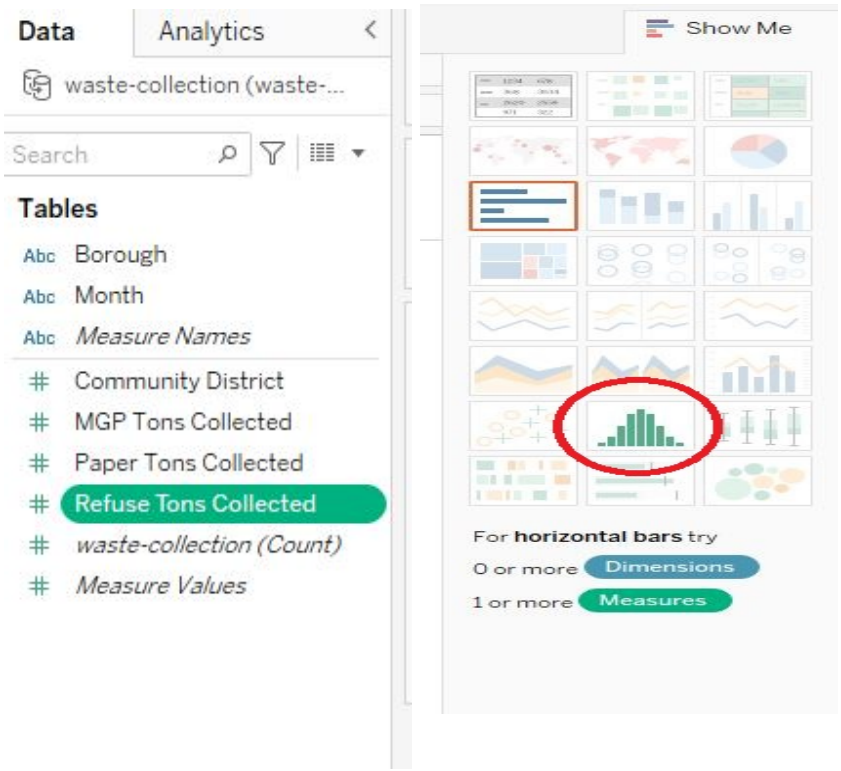
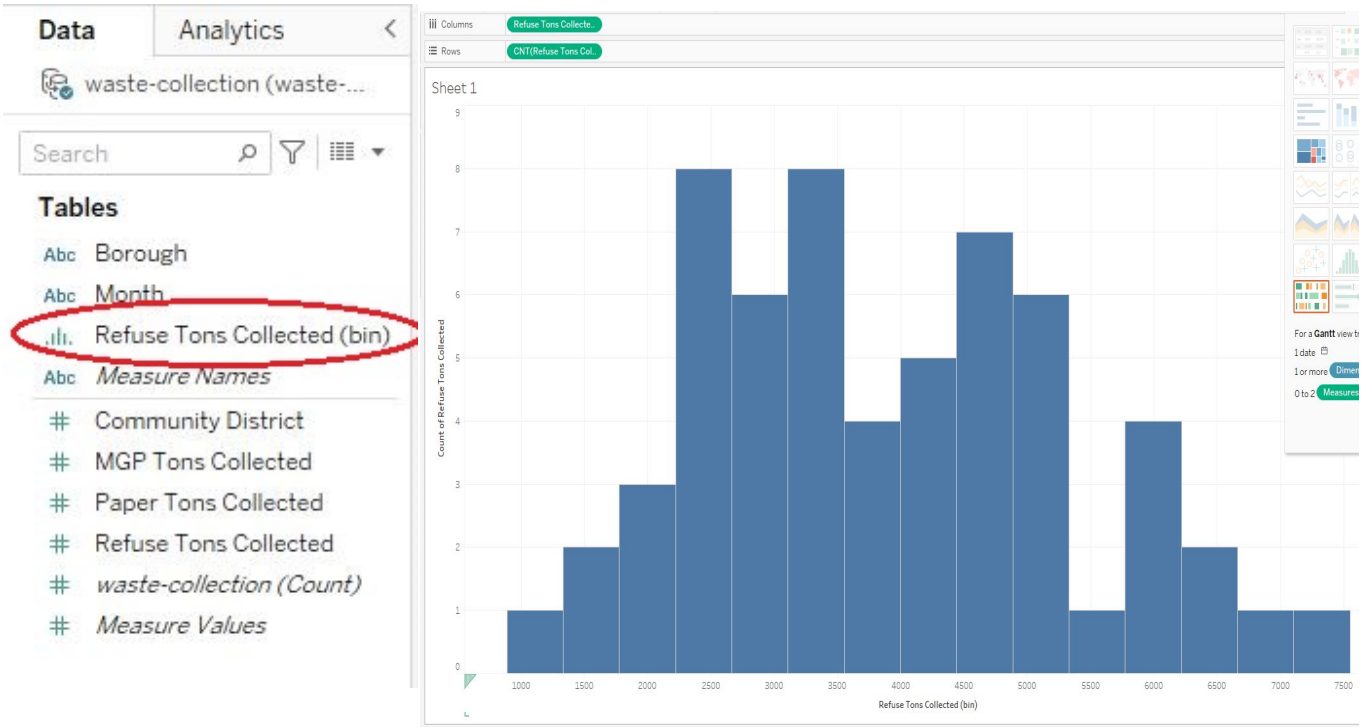


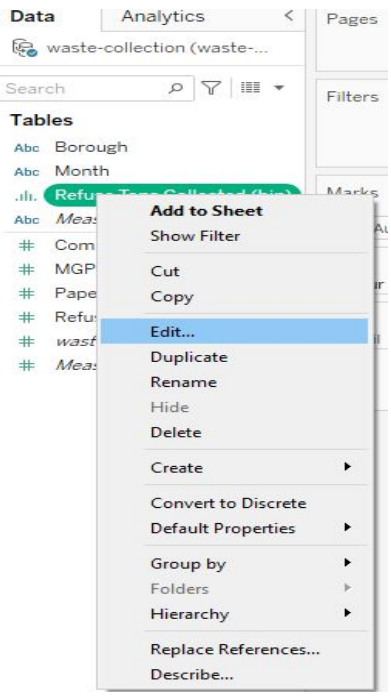
Tableau automatically creates a new field called **RefuseTonsCollected (bin)** in the dimensions area on Data pane. This data field is used to determine the discrete intervals into which the individual records are bucketed. The Rows shelf called CNT(RefuseTonsCollected) is used to determine the height of each bar. This field gives a value according to the number of records in each bin.

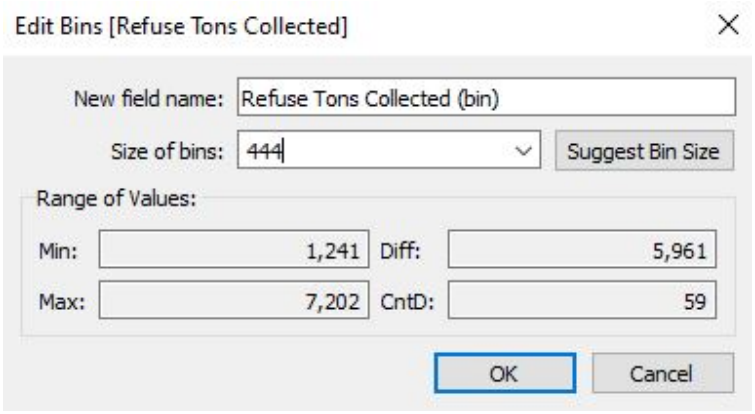


From this histogram, we can find the distribution of amount of refuse to be collected. The most common amount of refuse to be collected was between 2500 and 3500 tons (eight community districts fell in that range).

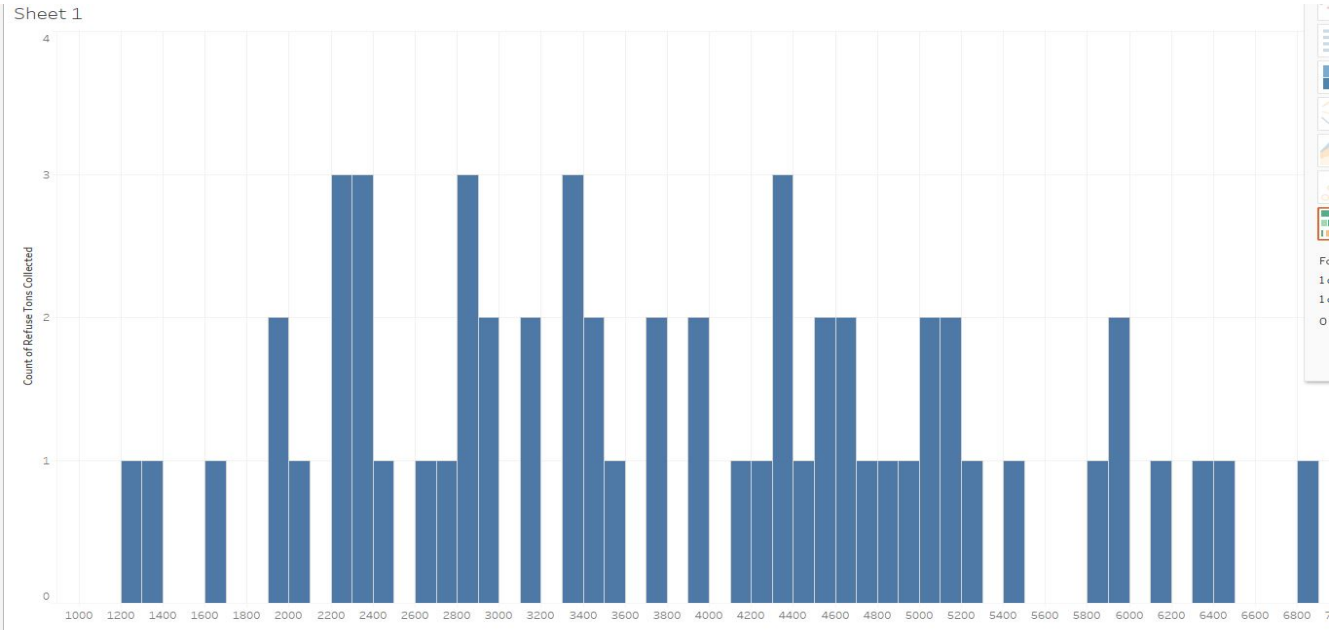
Chang the size of bins

We can modify the bin size by right-clicking on the newly created RefuseTonsCollected (bin) and then selecting Edit.





Change the bin size to 100, we'll get:



Project 5. Answering questions (Please do this at your home by using your own computer)

1. **Use Tableau to create a pie chart.**

Project 6. Create a Professional Report

Summarize the above experiments procedure, results, answering for questions and screenshots (project 1, 2, 3, 4, 5) into one report. Your report is the assignment that is required to be submitted for evaluation on week 8. Create a report by following the below steps.

Please note that

- as you may have a different version of Microsoft Word, the **following steps are just for your references.**

You can add a chapter called Chapter 5 in your previous report.

1. Open your last week's report and find the end of last week's report.
2. Copy this week's related experimental results, your findings and Screenshots, and paste them at the end of last week's report.
3. Delete original Table of Content you created.
4. Select all content, align all text to both left and right margin
5. Use shortcut key approach, generate Chapter 5: **Simple data visualizations**
6. Then use shortcut key approach to generate proper sub-chapters for this week's lab work.
7. Insert Table of Contents to your report.